

Evaluating the Environmental Impact of Low-Cost TLUD Pyrolysis Stoves on Smallholder Coffee Farms in the Ecuadorian Amazon

Abstract:

The Ecuadorian Amazon, a region renowned for its rich biodiversity, is home to thousands of small-scale coffee farmers who contribute significantly to the rural economy. However, this area faces environmental pressures due to deforestation and climate change. In response, biochar—produced through low-emission pyrolysis of agricultural waste—has become a promising tool for enhancing soil fertility, reducing greenhouse gas emissions, and sequestering carbon. This study investigates the potential of converting coffee husks into biochar using low-cost Top Lift Updraft (TLUD) stoves, a method that remains underexplored compared to traditional pyrolysis systems. The goal is to assess both the environmental and agricultural benefits of this approach, offering an alternative that is accessible to small-scale farmers. Six coffee farms with similar soil type, elevation, and climate will be selected. Three will utilize TLUD stoves to produce and apply biochar to their fields, while the remaining three will continue to practice traditional methods of husk disposal. Over six months, data will be collected monthly on visible emissions and soil quality indicators such as pH, nitrogen, and phosphorus levels. Emissions will be measured through qualitative estimation (e.g., color, opacity, duration, and smell intensity), and biochar production will be measured by mass. The results are expected to show that farms using biochar will experience improved soil health and reduced emissions. This research lays important groundwork for scaling affordable pyrolysis solutions in rural agricultural communities, aiding small farmers to use crop byproducts as a valuable resource rather than a pollutant.

Keywords:

Pyrolysis, Biochar, TLUD (Top Lift Updraft) Stoves, Smallholder Farming, Sustainable Agriculture, Agricultural Waste Management, Greenhouse Gas Emissions

Scientific Background:

Within its diverse ecosystems, the Ecuadorian Amazon plays a key role not only as a carbon sink but also as a home to thousands of smallholder farmers whose livelihoods depend on crops like coffee. In Ecuador alone, around 115,000 families work on smallholder coffee farms.¹ However, increasing temperatures, unsustainable land management, and deforestation have jeopardized the long-term sustainability of coffee farming in the Amazon.² In response, biochar—a form of charcoal produced by the burning of organic waste in low-oxygen conditions—has gained attention in the region for its potential to improve soil fertility, lock away carbon, and offer a cleaner way to manage agricultural residues. This is done using pyrolysis techniques, which often require a high initial cost to implement.³ Pyrolysis is the anaerobic decomposition of material by heating it; it produces biomass like biochar, bio-oil, along with gaseous byproducts.^{4,5} Investigating the environmental impacts of such stoves is essential to support sustainable agricultural practices and the resilience of local farming communities.

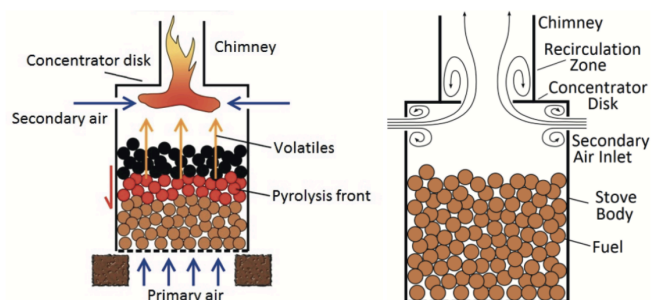


Fig. 1. *TLUD Stove Diagram*. Airflow, pyrolysis front, and chamber zones are labeled to illustrate the combustion process in a Top-Lit Updraft stove. Adapted from “*Design Improvements for Top-Lit UpDraft Biochar-Producing Gasifier Stove in Rural Kenya from the Users’ Perspective.*”⁶

Hypothesis:

If smallholder coffee farms in the Ecuadorian Amazon adopt low-emission stoves to convert coffee husks into biochar, then they will reduce pollutant emissions and enhance soil health more effectively than traditional coffee husk disposal methods.

Rationale:

While previous studies have explored the conversion of coffee husks via biochar, there is limited research on its use in rural settings like Ecuadorian coffee farms. Biochar, a byproduct of biomass pyrolysis, has shown potential to improve soil fertility and reduce greenhouse gases.^{7,8} Given the environmental vulnerability of this region, using low-cost, sustainable waste-to-energy systems, such as the TLUD stoves, is both urgent and potentially transformative. This study addresses a key knowledge gap in the agricultural sector and provides a nuanced understanding of energy production in the Ecuadorian Amazon.

Methodology:

Six smallholder coffee farms will be selected based on elevation, soil type, and climate to reduce variability. Three of these farms will utilize low-emission pyrolysis stoves (TLUD) to produce and apply biochar. The remaining farms will serve as control groups and will continue to use traditional practices; these will be grouped into Group A (open burning), B (composting), & C

(discarding).

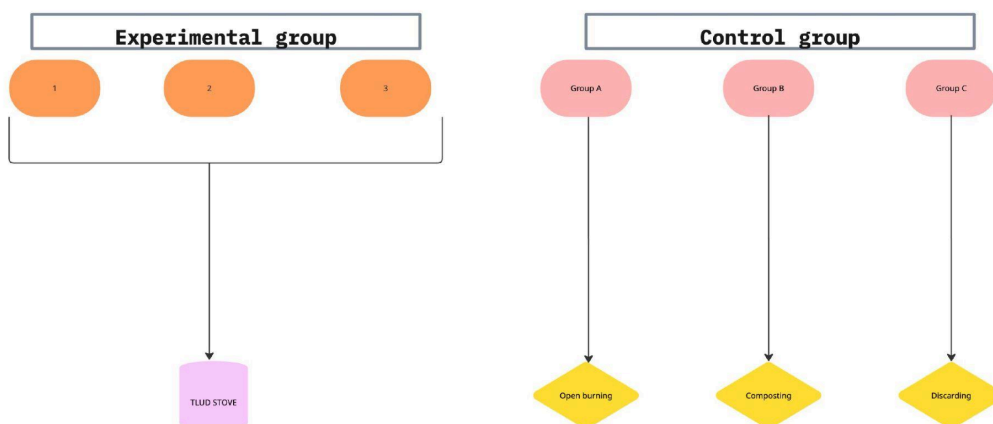


Figure 2: Experimental vs. Control Group

Starting measurements of nutrient quality and emission levels will be taken before and monitored monthly over 6 months. Each experimental farm will be provided with a TLUD stove, low-cost (built from local materials) and specifically designed for biochar production. Farmers will receive safety training on proper stove use. Following production, the resulting biochar will be applied to each farm. Soil and air samples will be collected, and data will be analyzed by comparing changes in environmental indicators using One-Way ANOVA.⁹ Changes in nutrient quality, such as pH, nitrogen, and phosphorus levels, will be measured. Emissions will be noted using qualitative traits (visible smoke level, burn time, and smell). All data collected will be compared across both treatment and control groups.

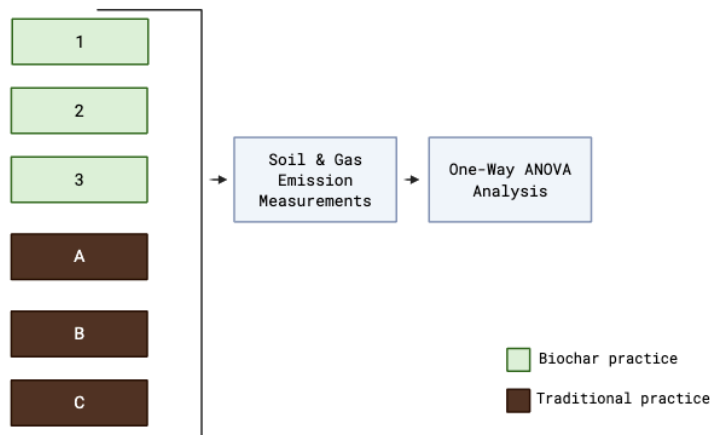


Figure 3: Experimental Design: Biochar vs. Traditional Practice. Recorded soil & gas emission measurements will be analyzed using one-way ANOVA analysis.

Farm	Treatment	Avg Soil pH	Avg Nitrogen %	Avg Phosphorus (mg/kg)
1	biochar	5.9	0.25	20.2
2	biochar	6.0	0.26	19.7
3	biochar	5.8	0.24	20.1
A	traditional	5.2	0.18	15.0
B	traditional	5.1	0.17	14.8
C	traditional	5.3	0.19	15.2

All values in the dataset are hypothetical, adapted from Sánchez-(2023) and Abbruzzini et al. (2023)

Figure 4: Hypothetical Result Table. Data will be recorded and compared after the 6-month trajectory. *Adapted from Sánchez-Reina et al. (2023) and Abbruzzini et al. (2023)*^{7,8}

Conclusion:

In hopes of supporting sustainable smallholder coffee farms in the Ecuadorian Amazon, this research was conducted to analyze the environmental impacts of TLUD stoves. The study

suggests TLUD stoves are a low-cost solution for smallholder farmers, offering improvements in soil quality and energy use.^{10,11} Grouping farms by traditional practices strengthened the comparisons, and combining field observations with data made the results more applicable. However, several limitations affect the generalizability of this study. These include the sample size, limited geographic scope, reliance on visual emission estimates, short monitoring timeframe, and unpredictable weather patterns. Future research should involve more farms and longer monitoring to assess long-term impacts. It should also incorporate quantitative emission tools and weather tracking data (e.g., rainfall, humidity, temperature) to better isolate the influence of stove use from climate variability. Even though the study was completed using a heuristic approach, it is an important first step in determining whether these technologies can be quickly and effectively applied to smallholder farms. This can help small coffee farmers in Ecuador dispose of their agricultural waste (in the form of coffee husks) and turn it into usable energy that is safe and environmentally friendly. The success of TLUD stoves in Ecuador may encourage adoption in other tropical agricultural regions with similar challenges.

References:

1. Ecuadorian coffee - adding flavour to the national economy | Rikolto in West Africa [Internet]. Rikolto.org. 2016 [cited 2025 Jul 27]. Available from: <https://westafrica.rikolto.org/id/node/3293>
2. Flores BM, Montoya E, Sakschewski B, Nascimento N, Staal A, Betts RA, et al. Critical transitions in the Amazon forest system. *Nature* [Internet]. 2024 Feb 14 [cited 2025 Jul 27];626(7999):555–64. Available from: <https://www.nature.com/articles/s41586-023-06970-0>
3. Heredia Salgado MA, Säumel I, Cianferoni A, Tarelho LAC. Potential for Farmers' Cooperatives to Convert Coffee Husks into Biochar and Promote the Bioeconomy in the North Ecuadorian Amazon. *Applied Sciences*. 2021 May 21;11(11):4747.
4. Juan-Carlos Cobos-Torres, Luis-Holguer Idrovo-Ortiz, Cobos-Mora SL, Santillan V. Renewable Energies and Biochar: A Green Alternative for Reducing Carbon Footprints Using Tree Species from the Southern Andean Region of Ecuador. *Energies* [Internet]. 2025 Feb 20 [cited 2025 Jul 27];18(5):1027–7. Available from: <https://www.mdpi.com/1996-1073/18/5/1027>
5. Ukanwa KS, Patchigolla K, Sakrabani R, Anthony E, Mandavgane S. A Review of Chemicals to Produce Activated Carbon from Agricultural Waste Biomass. *Sustainability* [Internet]. 2019 Nov 6 [cited 2025 Jul 27]; Available from: https://res.mdpi.com/d_attachment/sustainability/sustainability-11-06204/article_deploy/sustainability-11-06204-v3.pdf

6. Saraswati S. Design Improvements for Top-Lit UpDraft Biochar-Producing Gasifier Stove in Rural Kenya from the Users' Perspective [Internet] [PDF]. [Uppsala University]; 2018 [cited 2025 Jul 27]. Available from:
<https://www.diva-portal.org/smash/get/diva2:1264317/FULLTEXT01.pdf>
7. Sánchez-Reinoso A, Cerón-Muñoz MF, Vallejo A, Gómez M, Cárdenas M. Biochar effects on soil properties and coffee yield. *Environ Sci Pollut Res*. 2023;30:70124–70138. Available from: <https://doi.org/10.1007/s42729-023-01208-4>
8. Abbruzzini TF, Siqueira DS, Souza IM, Vasques GM, Jantalia CP, Urquiaga S, et al. Biochar and tropical soils: Effects on N₂O emissions. *Front Energy Res*. 2023;11:1092637. Available from: <https://doi.org/10.3389/fenrg.2023.1092637>
9. Ponomarenko T, Ecaterina Resniova, Patricio A. Assessment of Energy Sustainability Issues in the Andean Community: Additional Indicators and Their Interpretation. 2022 Feb 1 [cited 2025 Jul 27];15(3):1077–7. Available from:
<https://www.mdpi.com/1996-1073/15/3/1077>
10. Kiggundu N, Sittamukyoto J. Pyrolysis of Coffee Husks for Biochar Production. *Journal of Environmental Protection* [Internet]. 2019 [cited 2025 Jul 27];10(12):1553–64. Available from: <https://www.scirp.org/journal/paperinformation?paperid=96815>
11. Heredia MA, Luís A. C. Tarelho, Saeumel I. Post-extractivism and Bioeconomy: An Experimental Analysis of Combustion and Pyrolysis Processes as Alternatives to Add Value to Agro-Residues (Coffee Husks) Generated in Farmer Cooperatives of the Ecuadorian Amazon. 2024 Aug 6 [cited 2025 Jul 27]; Available from:
https://www.researchgate.net/publication/383128915_Post-extractivism_and_Bioeconomy_An_Experimental_Analysis_of_Combustion_and_Pyrolysis_Processes_as_Alternative

s_to_Add_Value_to_Agro-Residues_Coffee_Husks_Generated_in_Farmer_Cooperatives
_of_the_Ecuad